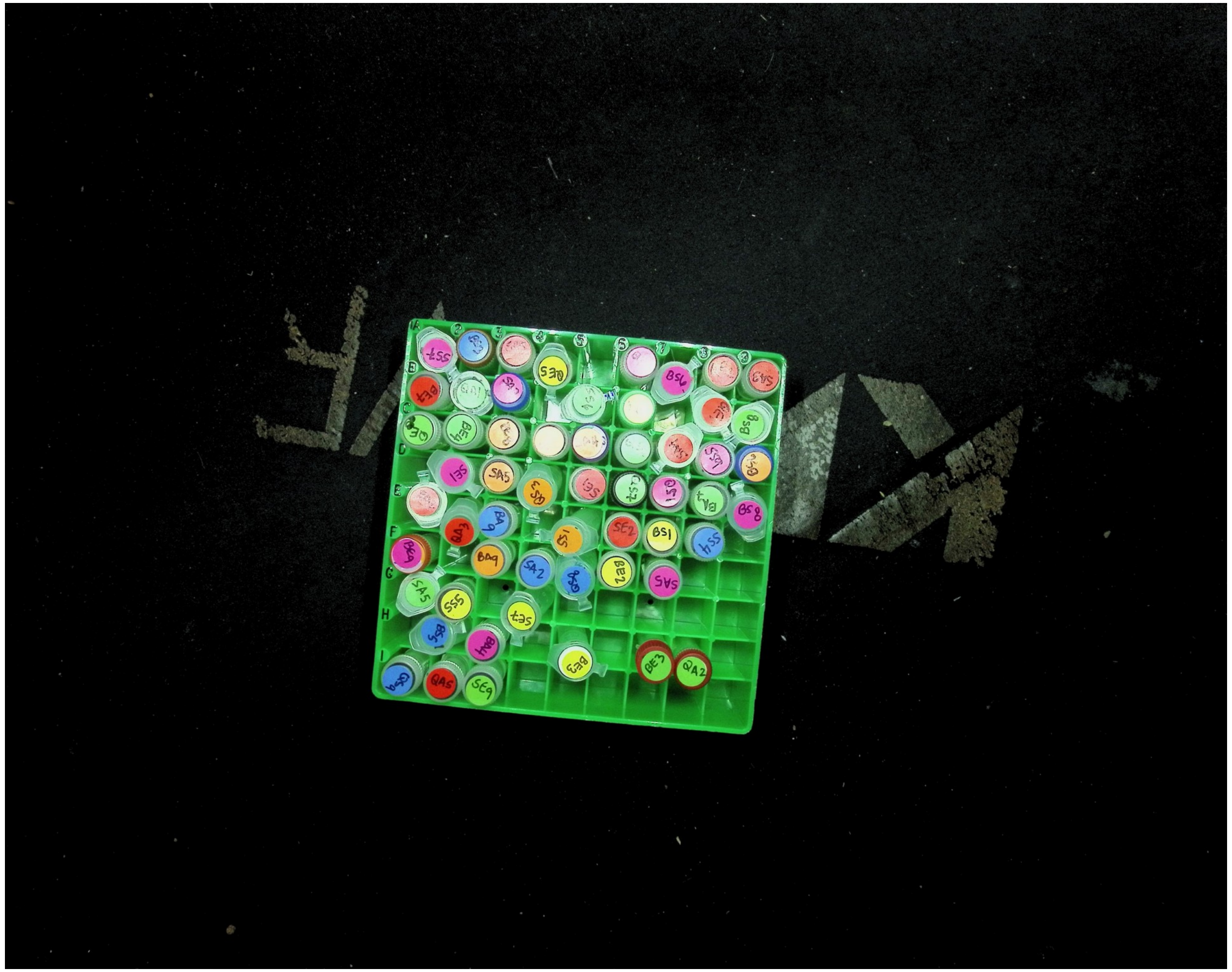
Roderick Taylor · Kate Leemann

LabRack Vision QA

YOLO11s/960 + OpenCV · staged, non-patient images · human review required

I support rack checks without replacing judgment



My prototype

I detect racks, caps, and possible empty slots in a still image.

My boundary

I flag possible issues for human review. I use no patient data and make no clinical claims.

What I built: one image becomes three artifacts

1

VALIDATE

JPG / PNG
path + extension

→

2

DETECT

fine-tuned
YOLO11s/960

→

3

REVIEW

counts +
cautious QA rules

→

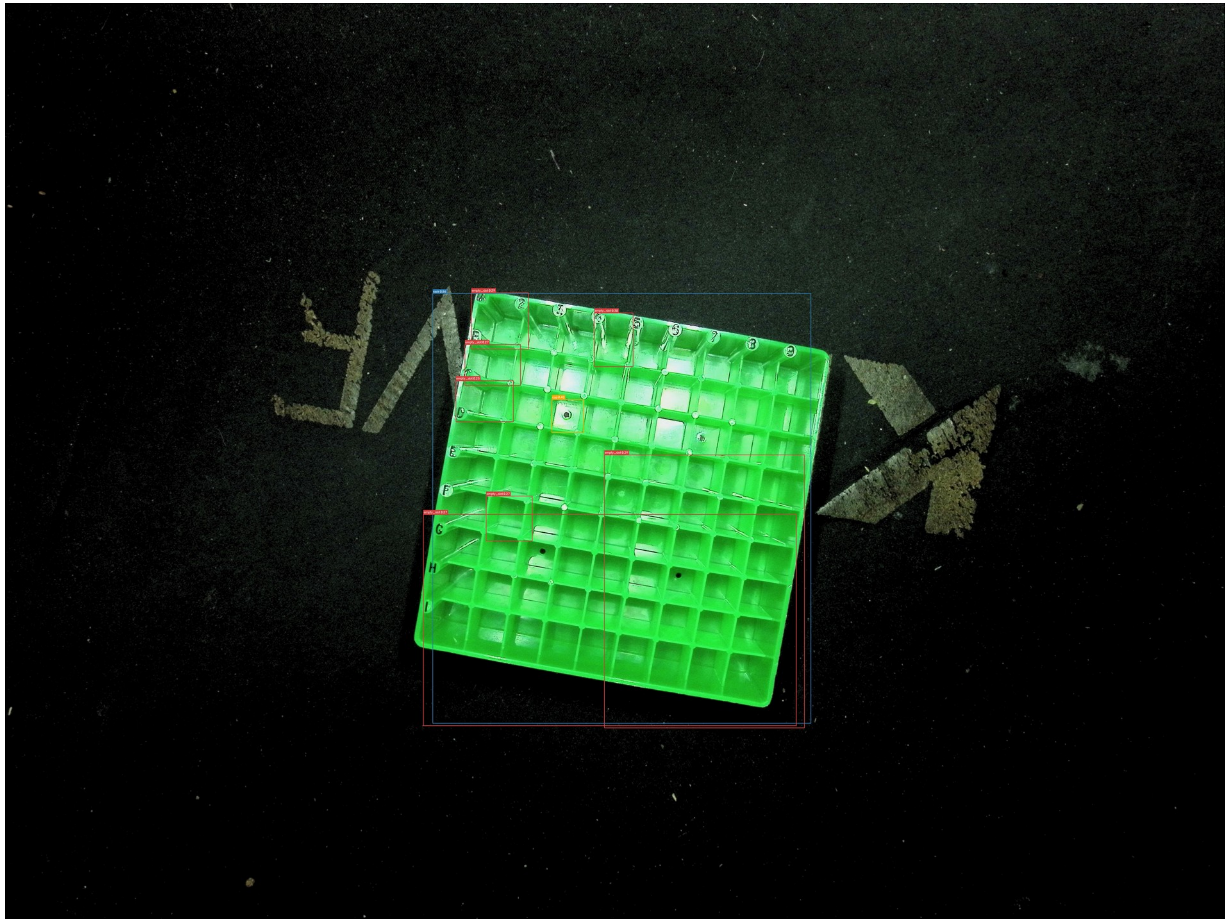
4

EXPORT

annotated image
JSON · summary

My fine-tuned detector feeds cautious QA rules and three reviewable outputs.

My demo produces image, JSON, and review output



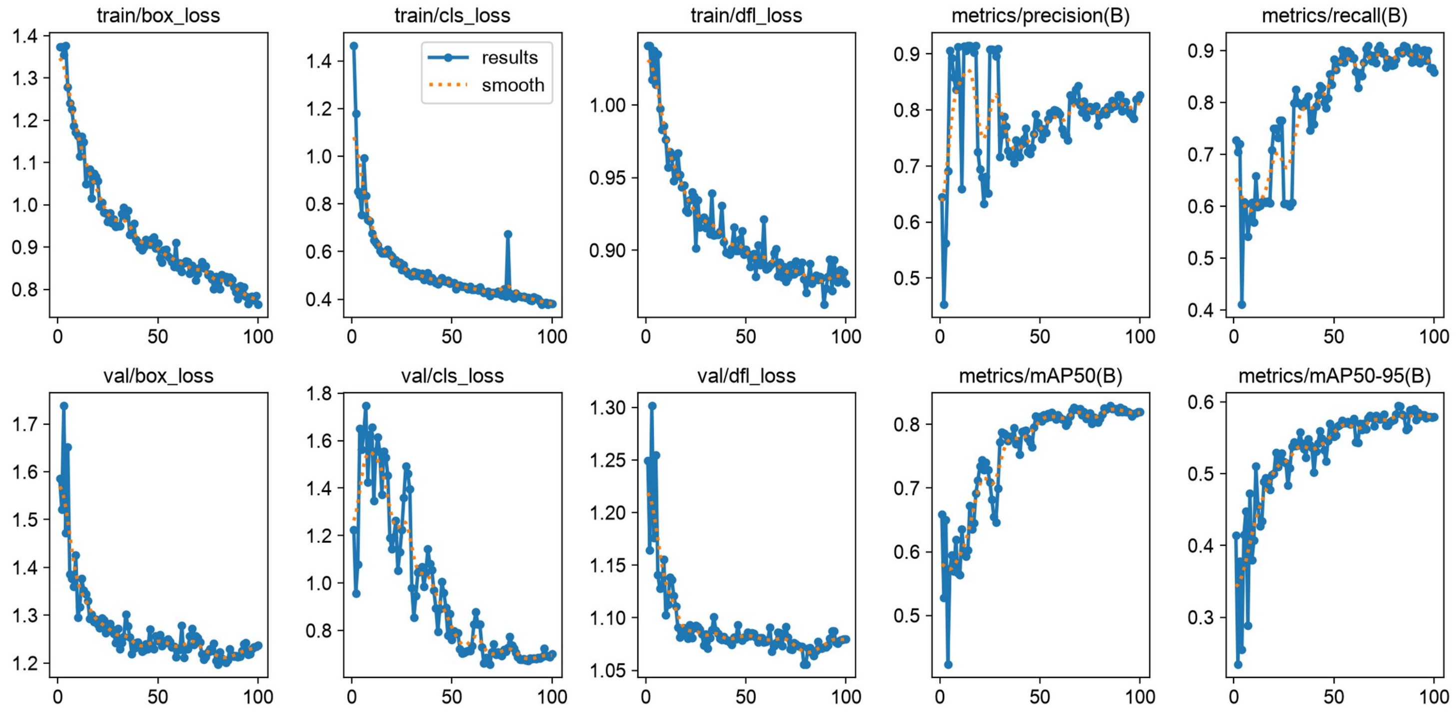
CLI SUMMARY

rack	1
tube	0
cap	1
empty_slot	7

8 detections below 0.50
→ manual review recommended

Detection step: 0.732 s

How I built it: YOLO11s transfer learning at 960 px



RUN

100 epochs
960 px · batch 2
YOLO11s

HARDWARE

Apple M3 Pro
PyTorch MPS

I observed MPS
non-deterministic-operation warnings.

I used a grouped split to keep rack images apart

140

staged images
partial Roboflow import

125 TRAIN · Rack_A/B	**5** VALIDATE · Rack_C	**10** TEST · Rack_D

Current annotation inventory

rack 133 cap 7,391 empty_slot 3,339

I, Roderick Taylor, handled programming + debugging.
https://www.linkedin.com/in/rodjtaylor/
Kate Leemann contributed photos + annotations and assisted me with testing methodology.
https://www.linkedin.com/in/kate-leemann-a21615236/

My results versus my Blueprint targets

0.937

target ≥ 0.75
test mAP50

2.07 s

target < 3 s
full CLI wall time

3 outputs

target: annotated
image + JSON + summary

Rack_D test: 10 staged images · 822 instances · mAP50–95 0.601

I met both numeric targets; my failure cases still require human review.

What I changed from my Blueprint

1

TECHNIQUE

segmentation planned
→ detection shipped

→

2

DATA

200–250 planned
→ 140 staged images

→

3

CLASSES

tube planned
→ not measured

→

4

MODEL

YOLO11 / seg planned
→ YOLO11s/960

I changed the scope to fit my staged data; I kept the end-to-end promise.

I made three fixes to get my prototype working

960 px

small empty slots
needed more detail

SPLIT

rack-level split
reduced leakage

BATCH 2

MPS smoke tests
proved the model fit

I changed resolution, grouping, and memory settings one at a time.

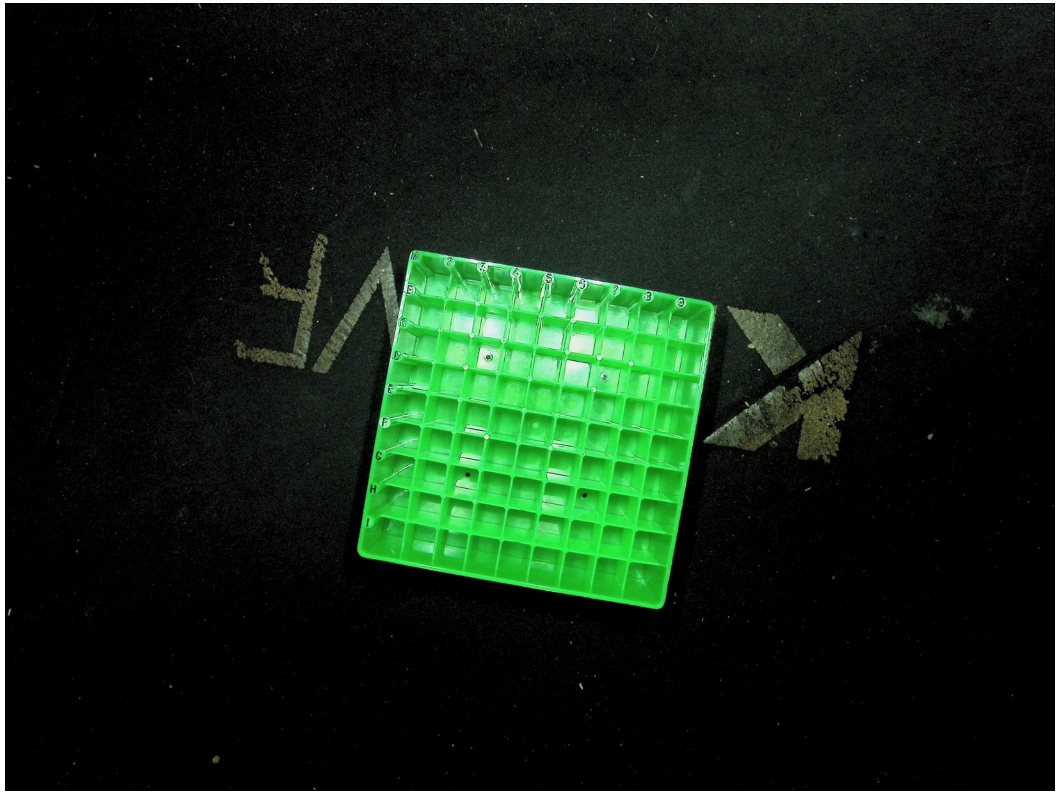
I expect slight MPS variation; every QA finding still requires human review.

My pipeline works; my dataset must grow.

1 I protect evaluation with grouped splits
2 I use 960 px for small slots
3 I need clearer staged tube images
4 I will add a new rack group + labels

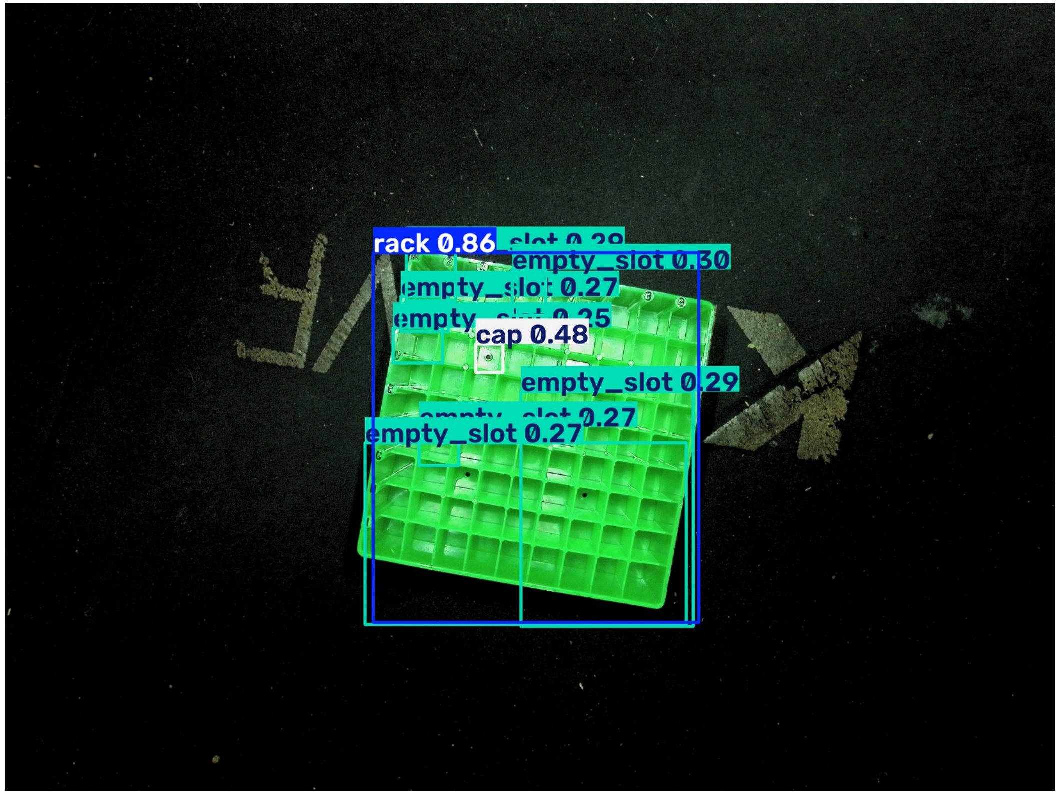
Human review required.
github.com/RTaylor89/labrack-vision-qa

My failure case: 7 of 81 empty slots detected



KATE'S ANNOTATION

81 empty · 0 caps



MY MODEL OUTPUT

7 empty · 1 cap